

Apolipoprotein B

Test Details

METHODOLOGY

Immunologic

RESULT TURNAROUND TIME

1 - 2 days

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

LDL and its major protein, apolipoprotein B, play an essential role in lipid transport and metabolism. Apo B may regulate cholesterol synthesis through its interaction with specific cell membrane receptors and by inhibition of HMG Co A reductase. This enzyme has been identified as the rate controlling enzyme in cholesterol biosynthesis. Apo B may be important in the genesis of atherosclerosis and its quantitation useful in the evaluation of patients at risk for or having coronary atherosclerosis (CAD). High levels indicate increased risk for CAD.

Specimen Requirements

Specimen

Serum (preferred) **or** plasma

Volume

2 mL

Container

Red-top tube, gel-barrier tube, green-top (heparin) tube, **or** lavender-top (EDTA) tube

Collection Instructions

Separate serum or plasma from cells. Transfer specimen to a plastic transport tube.

Stability Requirements

Temperature	Period
Room temperature	14 days
Refrigerated	14 days
Frozen	14 days
Freeze/thaw cycles	Stable x3

Reference Range

Apolipoprotein B (mg/dL)			
Desirable	Borderline High	High	Very High
<90	90 – 99	100 – 130	>130

ASCVD Risk Category	Therapeutic Target APO B (mg/dL)
Very High Risk	<80 (if extreme risk <70)
High Risk	<90
Moderate Risk	<90

Storage Instructions

Maintain specimen at room temperature.

Patient Preparation

Patient must be fasting 12 to 14 hours.

Causes for Rejection

Specimen from nonfasting patient

References

Colantonio DA, Kyriakopoulou L, Chan MK, et al. Closing the gaps in pediatric laboratory reference intervals: A CALIPER database of 40 biochemical markers in a healthy and multiethnic population of children. *Clin Chem*. 2012 May; 58(5):854-868. PubMed 22371482

(<http://www.ncbi.nlm.nih.gov/pubmed/22371482>)

Contois JH, McConnell JP, Sethi AA, et al. Apolipoprotein B and cardiovascular disease risk: position statement from the AACC Lipoproteins and Vascular Diseases Division Working Group on Best Practices. *Clin Chem*. 2009 Mar;55(3):407-419. PubMed 19168552

(<http://www.ncbi.nlm.nih.gov/pubmed/19168552>)

Jacobson TA, Ito MK, Maki KC, et al. National lipid association recommendations for patient-centered management of dyslipidemia: part 1–full report. *J Clin Lipidol*. 2015 Mar-Apr;9(2):129-169. PubMed 25911072

(<http://www.ncbi.nlm.nih.gov/pubmed/25911072>)